180Degrees
NITW

BREAK THE CASE

PRODUCT **ANALYTICS** **CONSULTING**

Rules & Guidelines

1. Project Selection

- There are four winter project questions, covering domains such as Data Analytics, Product, and Consulting.
- Participants may choose any ONE of the four questions.
- During registration, each team must select the specific question they will work on.
- 3rd and 4th years are encouraged to pick Questions 1 or 2, as these are perfectly suited for building a strong portfolio for data analyst roles.

2. Team Structure

- Teams can have 1 to 4 members.
- Cross-year or cross-department teams are allowed.

3. Project Duration

- The project begins on 27th November.
- The final submission deadline is 22nd December (11:59 PM).
- Late submissions will not be evaluated.

4. Mentorship & Communication

- Each project domain will have dedicated mentors assigned.
- All teams must join the respective project groups for updates, discussions, and clearing doubts with mentors.
- Collaboration within the group is encouraged; plagiarism is not.

5. Submission Guidelines

- All deliverables should strictly follow the format specified in the question document.
- Submissions must be original work.
- Only one submission per team will be accepted.

6. Evaluation & Certification

- Every team that successfully submits a valid project by the deadline will receive a Certificate of Completion.
- The top-performing teams across all categories will be awarded a Certificate of Merit.

7. General Conduct

- Participants must maintain professional behaviour in mentor groups.
- Misconduct, plagiarism, or violating deadlines may lead to disqualification.

1 Building End-to-End Supply Chain Visibility Using SQL

Introduction

FreshCart Pvt. Ltd. is a fast-growing consumer goods company. It operates several factories, regional warehouses, distributors and retail outlets across India. The company sells more than 800 SKUs, including packaged foods, beverages, home care and personal care products.
As the company expands, supply chain operations are becoming more complex. FreshCart generates data from purchase orders, supplier deliveries, warehouse scans, shipments and sales. However, this data is not being used properly. Several inefficiencies continue to affect daily operations.

The Challenge

FreshCart currently faces the following problems:

- Frequent shipment delays and inconsistent On Time In Full performance
- Stockouts in some regions even when total national inventory is enough
- Excess inventory in slow markets that increases holding cost
- Lack of clarity about which suppliers are reliable
- Limited insight into warehouse capacity, movement speed and bottlenecks
- No single place where supply chain KPIs are tracked or monitored

The company wants to use SQL-based analytics to create a clear view of its end-to-end supply chain.
SQL will help convert raw datasets into meaningful insights and help improve performance.

Your Mission

As project participants, your task is to design and implement a SQL-based supply chain visibility system.
The goal is to help FreshCart understand what is happening at every stage of the supply chain.
You are expected to:

- Create SQL queries to extract, clean and analyse supply chain data
- Find inefficiencies in supplier performance, warehouse movement and transportation
- Generate supply chain KPIs through SQL
- Provide insights that FreshCart can act on.

Submit both technical outputs and your analytical recommendations

---

Key Tasks and Deliverables

SQL Queries

You will write and test SQL queries for:

- Fill Rate across warehouses and regions
- On Time In Full measurement for suppliers and transporters
- Order Cycle Time from purchase order creation to customer delivery
- Backorder identification
- Inventory Turnover and inventory ageing
- Warehouse inbound and outbound movement
- Delay patterns across routes and carriers

Database Optimization

You will also:

- Normalize the raw datasets into a clean relational structure
- Use SQL practices that improve performance such as indexing, joins, window functions and CTEs.
- Document your data cleaning and transformation decisions
- Create an ERD that shows how all the tables connect

Analytical Outputs

Using your SQL queries, you should prepare a report that:

- Identifies where the supply chain is slowing down
- Shows which SKUs move fast and which move slow
- Suggests inventory movement between regions
- Highlights unreliable suppliers or transport carriers
- Explains demand patterns

What We Expect From You

- Clean and well-structured SQL code
- Clear analytical thinking and problem solving
- Simple communication of your findings
- Effort to explore beyond the basic tasks if you see additional valuable insights

---

Expected Business Impact

By completing this project, your SQL system should help FreshCart:

- Understand the full supply chain in one place
- Reduce stockouts
- Lower excess inventory and holding cost
- Improve supplier and logistics performance
- Make better, data-based decisions
- Improve the reliability of product availability for customers

Final Submission Should Include

- SQL scripts and documentation
- ERD or schema diagram
- A small KPI report or dashboard
- A short executive summary explaining your insights and recommendations

---

Dataset

Dataset Link

Project Mentors

Bibek: 8092225219
Donna: 6361869830

Resource

Link

Whatsapp Group

Link

2 Analysing Urban Air Pollution Patterns

Introduction

Air quality has become one of the most urgent environmental challenges in many Indian cities.
Pollutants such as PM2.5, PM10, NO2 and SO2 frequently rise above recommended limits.
Factors such as vehicle traffic, construction dust, waste burning, industrial activity and seasonal changes all contribute to poor air quality.
Although air-quality readings are collected across multiple monitoring stations, this data is not analysed deeply or consistently. As a result, authorities and citizens often do not know which areas are most affected, what times pollution is highest, or what trends are emerging over time.
This project focuses on using Excel and Power BI to study air pollution data and present insights in a clear, data-backed manner.

The Challenge

Cities face several challenges while dealing with air pollution:

- Large differences in pollution between neighbourhoods
- Sharp increases during specific hours, especially traffic peaks
- Seasonal variations driven by weather and local activities
- Limited understanding of long-term patterns
- Low public awareness of high-risk times
- Insufficient data-driven planning by local authorities

Your task is to use data tools to understand these patterns and propose practical interventions.

Your Mission

As project participants, your goal is to conduct an end-to-end analysis using Excel and Power BI. You will explore the data, identify meaningful trends and create strategic recommendations that can help a city manage air quality better.
You are expected to:

- Clean and prepare the dataset using Excel
- Perform descriptive analysis to understand daily and monthly variation
- Build pivot tables, charts and summary insights
- Create an interactive Power BI dashboard
- Explain when and where pollution is worst
- Recommend practical actions based on your findings

Key Tasks and Deliverables

Data Preparation and Excel Analysis

Your Excel work will include:

- Cleaning the dataset.
- Converting raw readings into daily, weekly and monthly summaries
- Creating pivot tables by pollutant, location and time slot
- Identifying sudden spikes, dips and outliers
- Comparing pollution levels across stations and seasons
- Calculating averages, medians and peak-hour values

Power BI Dashboard

Your dashboard should include:

- Pollution level trends over time
- Location-wise comparisons
- Heatmaps or hotspot identification
- Hour-of-day and day-of-week patterns
- Seasonal variation summaries
- Simple forecasting using Power BI's built-in tools

The dashboard should be easy for a non-technical audience to interpret.

Strategy Recommendations

Based on your findings, prepare clear and actionable recommendations.
These may include:

- Identifying the most critical hotspots
- Explaining which pollutants require the most focus
- Proposing short-term measures like traffic control during peak hours
- Suggesting enforcement for construction dust and waste burning
- Highlighting priorities for public awareness
- Recommending long-term actions such as green corridors, public transport improvements and
- additional monitoring stations
- Your strategy should be realistic and supported by the data you analysed.

What We Expect From You

- Well-organised Excel files
- A neat, functional Power BI dashboard
- Clear explanations of patterns
- Recommendations that are logical and feasible
- Ability to present insights in a structured manner

Expected Project Impact

Your project will demonstrate how data analysis can support better environmental planning. The work will help:

- Identify high-risk areas and time periods
- Show how different pollutants behave over time
- Support city-level planning and resource allocation
- Improve public health communication
- Encourage evidence-based policy decisions
- The aim is to help authorities and communities take well-informed and effective actions.

Final Submission Should Include

Cleaned Excel workbook
Power BI dashboard file
A 10-page insight and recommendations report

Dataset

Dataset Link

Project Mentors

Akash: 8074549484
Om : 9579298602

Resource

Link

Whatsapp Group

Link

3

Designing a Food-Saving & Surplus-Redistribution App

180Degrees NITW

Introduction

Across many countries, apps like Too Good To Go help customers buy unsold surplus food from restaurants at discounted prices. This reduces food waste and helps businesses recover some value instead of throwing food away.
India currently does not have a simple and widely adopted version of this idea. Large amounts of fresh food still get wasted every day in restaurants, bakeries and grocery stores. At the same time, students, office workers and low-income consumers look for affordable meals every day.
This project focuses on designing a product that adapts this model for Indian users and local food businesses.

The Challenge

Key problems in India include:

- Restaurants and shops waste a lot of unsold food
- Customers want affordable options but do not know where to find surplus meals
- No single platform connects surplus food with buyers
- Businesses lose revenue at the end of the day
- Food waste contributes to environmental harm

Your challenge is to design a simple, practical app that solves these issues.

Your Mission

As participants, you will act as product managers and design a product concept called SaveBite, a platform that connects users with discounted surplus food nearby.
You are expected to:

- Understand user needs
- Define the product problem clearly
- List core features
- Design 2 to 3 key wireframes
- Create KPIs and success measures
- Prepare a short product case deck

Key Tasks and Deliverables

Problem and User Understanding

Define the main user groups:

- Customers searching for affordable meals
- Restaurants wanting to sell leftover food
- College students and office workers
- Optional: small bakery and café owners
- List the main pain points for each user group.

Product Scope and Feature Definition

Define the most important features:

- List of nearby surplus food options
- Simple discount-based pricing
- Pick-up time windows
- Quick payment
- Restaurant upload flow for surplus items
- Basic rating system
- Keep the scope realistic for an early product version.

Wireframing (2 to 3 Screens Only)

Create simple wireframes for these key screens:
Screen 1: Home Page showing nearby surplus food options
Screen 2: Food Item Details showing price, pick-up time and order button
Screen 3: Order Confirmation with pick-up instructions
These three screens should represent the core user journey from discovery to pick-up.

KPIs and Success Metrics

Possible KPIs include:

- Surplus food sold per day
- Restaurant participation and activity
- Repeat purchases
- Average discount claimed
- Daily active users
- Pick-up completion rate
- Customer satisfaction rating
- Choose the KPIs that best fit your product vision.

Go-To-Market and Strategy

Explain:

- Your launch target (students, working professionals, urban neighbourhoods)
- City or locality where you will start
- Partnership ideas (bakeries, cloud kitchens, small restaurants)
- Marketing ideas for early adoption
- A three-month feature roadmap
- Keep the strategy practical and easy to execute.

Product Case Deck

Your deck should include:

- Problem
- User personas
- Feature list
- 2 to 3 wireframes
- Product flow
- KPIs
- Launch strategy
- Risks and assumptions
- Conclusion

What We Expect From You

- Clear explanation of the problem
- Simple and practical product thinking
- Wireframes that show the core flow
- KPIs that match the product idea
- A clean and structured case deck

Expected Project Impact

Through this project, you will learn:

- How to think like an early-stage product manager
- How to define and measure success
- How to design simple wireframes
- How to create a realistic product strategy
- How to communicate product decisions clearly

These skills are essential for PM internships and early product roles.

Final Submission Should Include

- User personas
- Feature scope
- 2 to 3 key wireframes
- KPIs
- A structured 15 slide case deck (including Intro & appendix)
- A short summary note

Dataset

Not required.

Project Mentors

Abhinay: 9182796023
Rishi : 7840928905

Resource

Link

Whatsapp Group

Link

4

Market Entry & Feasibility Strategy for an EV Battery-Swapping Network in India

180Degrees NITW

Introduction

India's electric two-wheeler market is growing rapidly, driven by delivery riders, gig workers and urban commuters. But slow charging, long wait times and range anxiety continue to be major barriers.

Battery swapping has emerged as a promising alternative. It allows riders to exchange a discharged battery for a fully charged one within minutes. While this model has succeeded in some countries, India still does not have a large, reliable and profitable swapping network. A global energy-tech company is exploring an entry into India with a dedicated battery-swapping network. Before investing, the company needs a clear understanding of whether this opportunity is attractive, viable and scalable.

The Question

Should the company enter India's EV two-wheeler market with a battery-swapping network?

If yes, which cities should it begin in, what business model should it follow, and how should it expand over the next few years?

Your task is to analyse the opportunity, evaluate feasibility and recommend a market-entry and growth strategy.

Scope of Work

Participants are expected to explore the following:

- Understand the EV two-wheeler landscape in India
- Analyse demand drivers, industry trends and customer behaviour
- Benchmark existing players and their models
- Study state-level EV incentives and regulatory factors
- Identify the most suitable cities for a pilot
- Define a practical business model and pricing approach
- Outline operational needs such as station network and battery pools
- Explore potential partnerships with OEMs, delivery platforms and fleet operators
- Estimate the investment required and expected economics
- Create a simple roadmap for expansion
- Evaluate risks and provide a clear recommendation

Your work should be structured, logical and supported by data.

Key Deliverables

Your final submission should include:

1. **Problem understanding**

A short explanation of the challenge and opportunity.

2. **Market and city assessment**

Key trends, insights and city selection rationale.

3. **Competitor and policy overview**

What others are doing and key regulatory requirements.

4. **Business and pricing model**

How the company should operate and make money.

5. **Financial feasibility**

Basic economics, cost drivers and potential profitability.

6. **Operations and infrastructure plan**

How the swapping network would function on the ground.

7. **Digital platform concept**

2–3 core screens or system components.

8. **Go-to-market and expansion roadmap**

Steps for launching and scaling.

9. **Risks and final recommendation**

Top risks and your final answer to the main question.

Final Submission Should Include

- A 18-25 slides case deck(including Introduction and Appendix)

Dataset

Not required.

Project Mentors

Mridula : 9121336981
Disha : 9353476505

Resource

Link

Whatsapp Group

Link